

LESSON PLAN	COURSE: MDM4U
Date:	Unit of Study: Unit 2
Materials/Resources: Mathmatics of data management study guide	Lesson Focus/Theme: Chapter 4: Permutations and organized counting
Prior Knowledge: Data analysis	
Overall Expectations: By the end of this lesson, students will: solve problems involving the application of permutations and combinations to determine the probability of an event.	
Specific Expectations: 2.1 recognize the use of permutations and combinations as counting techniques with advantages over other counting techniques (e.g., making a list; using a tree diagram; making a chart; drawing a Venn diagram), distinguish between situations that involve the use of permutations and those that involve the use of combinations (e.g., by considering whether or not order matters), and make connections between, and calculate, permutations and combinations 2.2 solve simple problems using techniques for counting permutations and combinations, where all objects are distinct, and express the solutions using standard combinatorial notation [e.g., $n!$, $P(n, r)$, $(n\ r)$]	
Learning Goals: By the end of this lesson, students will know the organized counting, factoirals and permutations	

Success Criteria:

By the end of this lesson, I will be able to do organized counting

By the end of this lesson, I will be able to do the factorials and permutations

Assessment Strategies: Tests, quiz and project**Time:**
10 min**Introduction/Minds-On:**

Answer questions students don't understand from last class if any

Time:
90 min**Lesson:**

Chapter 4: Permutation and organized counting

4.1 Organized counting

4.2 Factorial and permutations

Time:**Extension / Wrap-up:**

10 min

Review the key concepts and formula for this lesson

10 min

Review for test

Notes / Reminders / Homework:

Do test #1 before or after the class in classroom for one hour

Lesson Reflection:

